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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 21: EIA Case study - V

By-
Prof. Bhola Ram Gurjar
Dept. of Civil Engineering

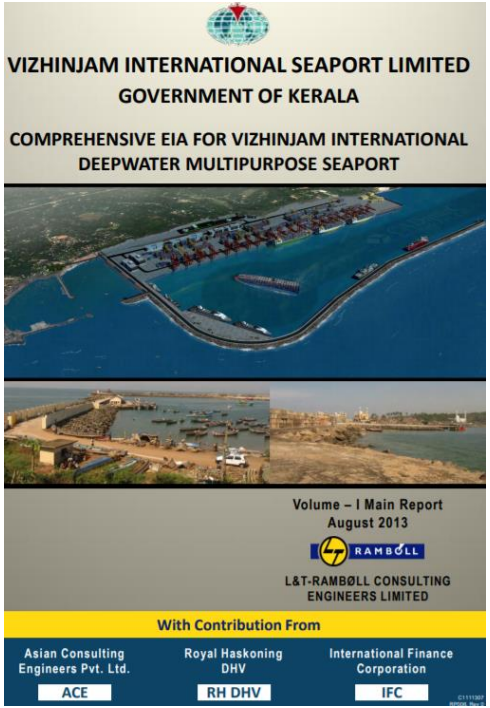


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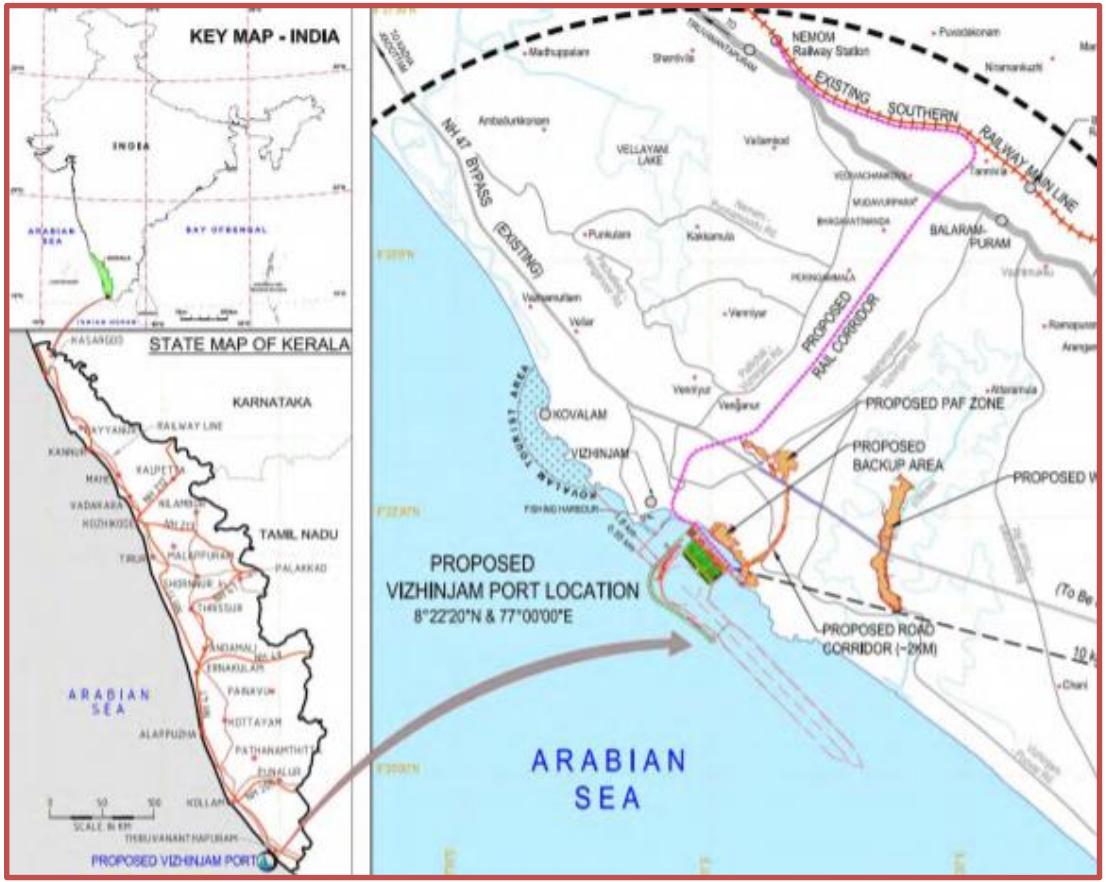
- EIA for Vizhinjam international deep-water multipurpose seaport (in state of Kerala, India)
  - Project proponent & lender
  - Benefits of project
  - Phases of project & its timeline
  - Environment clearance
  - Alternative sites considered for project
  - Comparison of alternative sites
  - Major impacts & mitigation measures
  - Public participation
  - Environment management plan

Comprehensive EIA for Vizhinjam international deep-water multipurpose seaport

- Project: Vizhinjam international deep-water multipurpose seaport
- Location: Vizhinjam, Kerala (India)
- Project proponent: Vizhinjam International Seaport Limited (VISL)
- EIA report by: L&T-RAMBOLL Consulting Engineers Limited



Project location on map



Project location

- The project site is located at Vizhinjam, Thiruvananthapuram District, Kerala

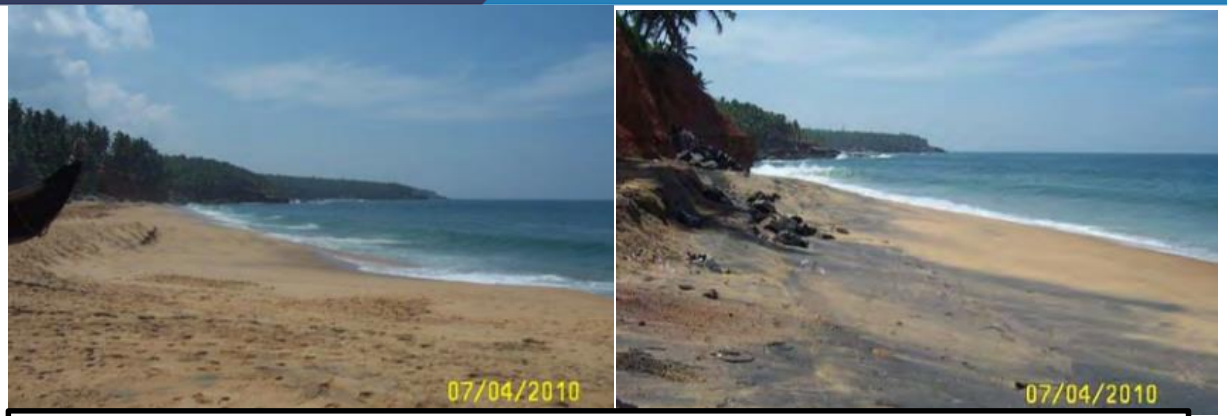


Image: Adimalathura (Sothern End) & Nellikkunnu (Northern end)

- The port facilities (including quays, terminal area and port buildings) are proposed to be developed entirely on reclaimed land
- Adequate backup area and other land resources have been acquired in the vicinity of proposed port

Key words related to port projects

- Reclaimed land: Land obtained though filling any marshy land or seabed
- Quays: Platform on ports for loading-unloading of goods
- Backup area for ports: Area within port premises for parking, offices, storage etc.
- Contours: Lines representing same height or depth on map
- Greenfield project: A new project starting from scratch

1 Nautical mile= 1.852 km

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Vizhinjam International Seaport Limited (VISL)

- Vizhinjam International Seaport Ltd - a company fully owned by Government of Kerala
- It is the implementing agency for the project, will be responsible for all obligations and responsibilities of the Government of Kerala
- Exclusive new organization for implementing project



International Finance Corporation (IFC)

- IFC is an international funding organization; part of world bank group
- Works towards the World Bank Group's twin goals of ending extreme poverty and boosting shared prosperity
- Advances economic development and improves the lives of people by encouraging the growth of the private sector in developing countries



Major site-specific advantages

- Availability of 18-22 m contour within one nautical mile from the coast
- Proximity to International shipping routes approximately 10 nautical miles
- Minimal drift along the coast
- Hardly any maintenance dredging is required
- Links to National/regional road, rail
- Flexibility in design and expansion being a greenfield project



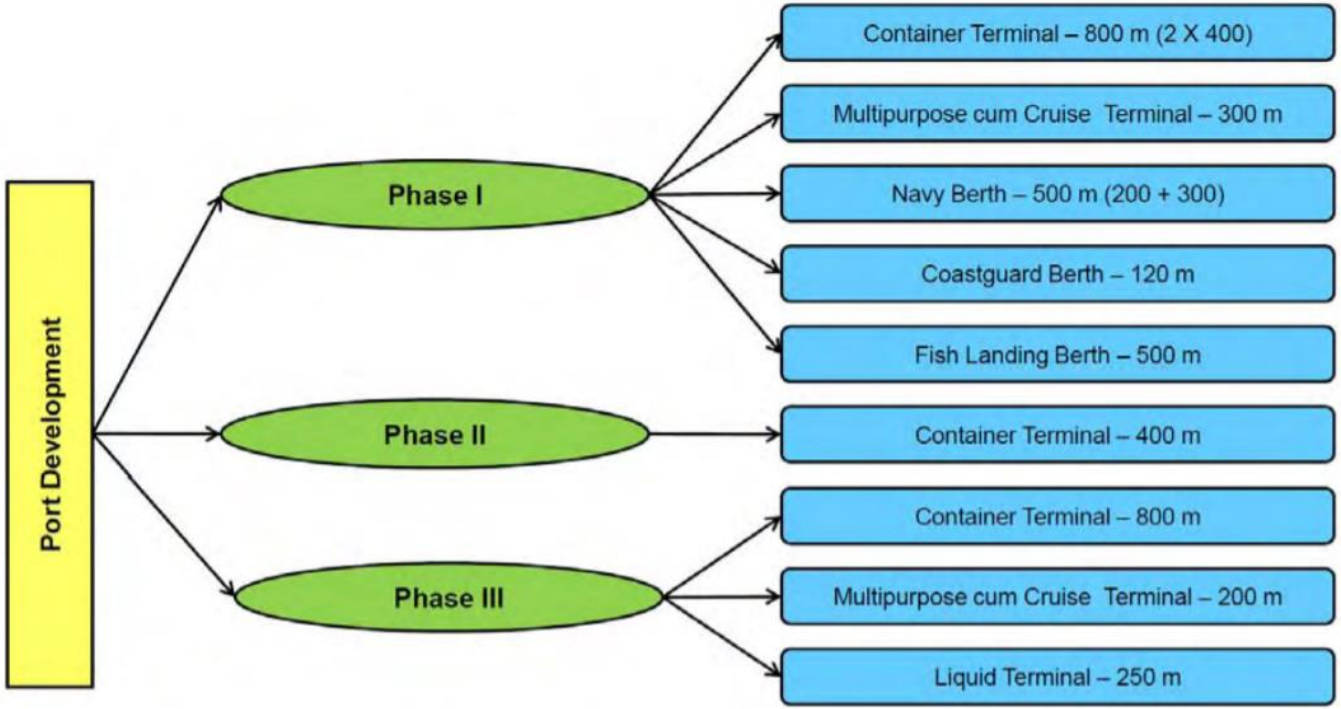
Major benefits of project 1/2

- Improved Socio-Economic Conditions
- Employment potential during the construction phase - 2350 persons
- Expected direct employment during operation phase - 650 persons
- 2000 persons of indirect employment
- Strategic defense Location
- Major Boost to South Indian Tourism Industry
- Project Benefits due to Revenue Generation

Major benefits of project 2/2

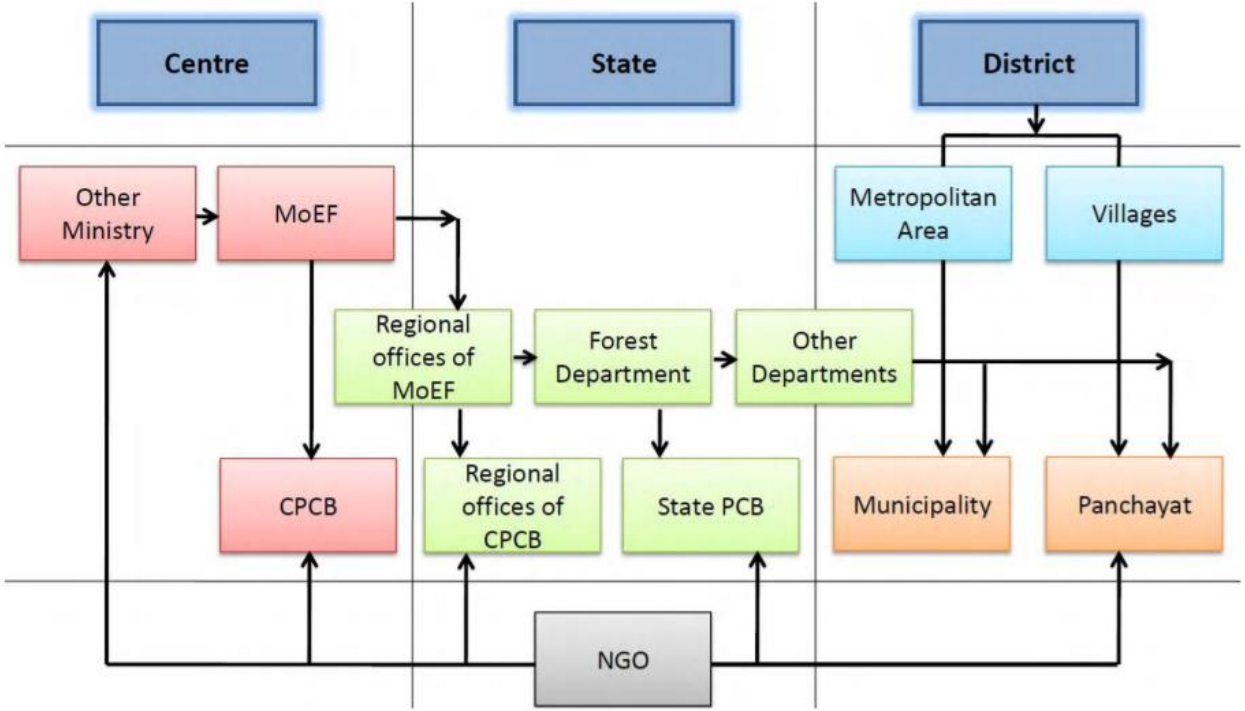
- Development of institutions like schools, colleges and offices
- Benefits to agricultural sector
- Infrastructural facilities
- Biodiversity and Climatic benefits
- Induced development
- Major boost to southern India’s tourist industry – Cruise terminal and tourism
- Improved Socio-economic conditions

Phase-wise development of the proposed Port



Phase I : 2014-17
Phase II : 2024-27
Phase III : 2034-37

Overview of framework for Environment Clearance: India



MOEF&CC: Ministry of Environment, Forest & Climate Change

CPCB: Central Pollution Control Board

Indian acts & regulations for environment protection

- Environment (Protection) Act 1986
- Air (Prevention and Control of Pollution) Act and Rules 1981
- Water (Prevention and Control of Pollution) Act and Rules 1974
- The Indian Forest Act 1927

- Biological diversity act 2002
- EIA notification 2006
- Coastal regulation zone notification 2011
- Noise pollution (control & regulation) rules 2000
- Solid waste management rule 2000

- National resettlement & rehabilitation policy 2007
- Public liability act 1991

Requirement from International Finance Corporation (IFC)

Performance Standards (PS) on Social and Environmental Sustainability by IFC are given as:

PS	Details
PS 1	Social and Environmental Assessment and Management System
PS 2	Labor and Working Conditions
PS 3	Pollution Prevention and Abatement
PS 4	Community Health, Safety and Security
PS 5	Land Acquisition and Involuntary Resettlement
PS 6	Biodiversity Conservation and Sustainable Natural Resource Management
PS 7	Indigenous People
PS 8	Cultural Heritage

Being part of World bank, EIA requirements of IFC are same as World Bank

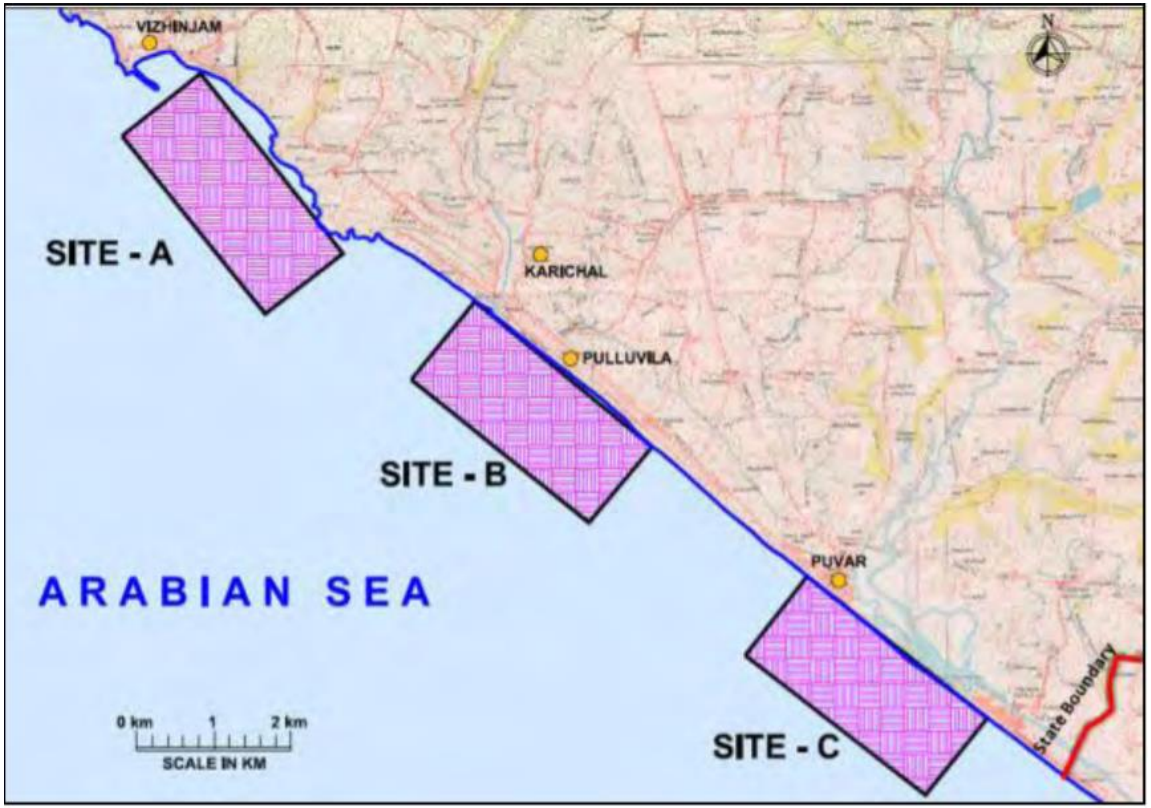
Influenced area/General study area

- Area within 10 km radius with the Vizhinjam port site as a center has been earmarked for the study as the general study area
- 30 km stretch (15 km both sides) of the shoreline was studied for social impacts and shoreline impacts
- Core study area is the acquired project area with a two-kilometre radius from the center of the port site

Phase-I project cost

Description	Phase I (in ₹ Crores)
Port Facilities (VISL + Private Operator)	3640 (2332 + 1308)
Road	60
Rail	350
Power	50
Water	10
CSR & EMP Costs [5% of 2332 crores (VISL portion of phase I port costs) plus 5% of 470 crores external infrastructure cost)]	140
Land Cost (140.42 Ha)	937
Total	**5,187**

Alternatives for site for project

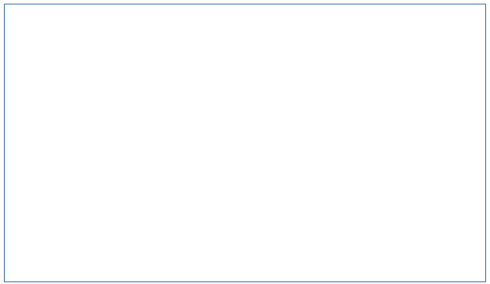


Site	Location
A	Vizhinjam
B	Pulluvila
C	Poovar

Criteria for site selection

Following criteria/considerations have led to the selection:

- Port Development
- Operational & Navigational
- Connectivity with other modes of transportation
- Environmental Aspects
- Social Aspects
- Cost



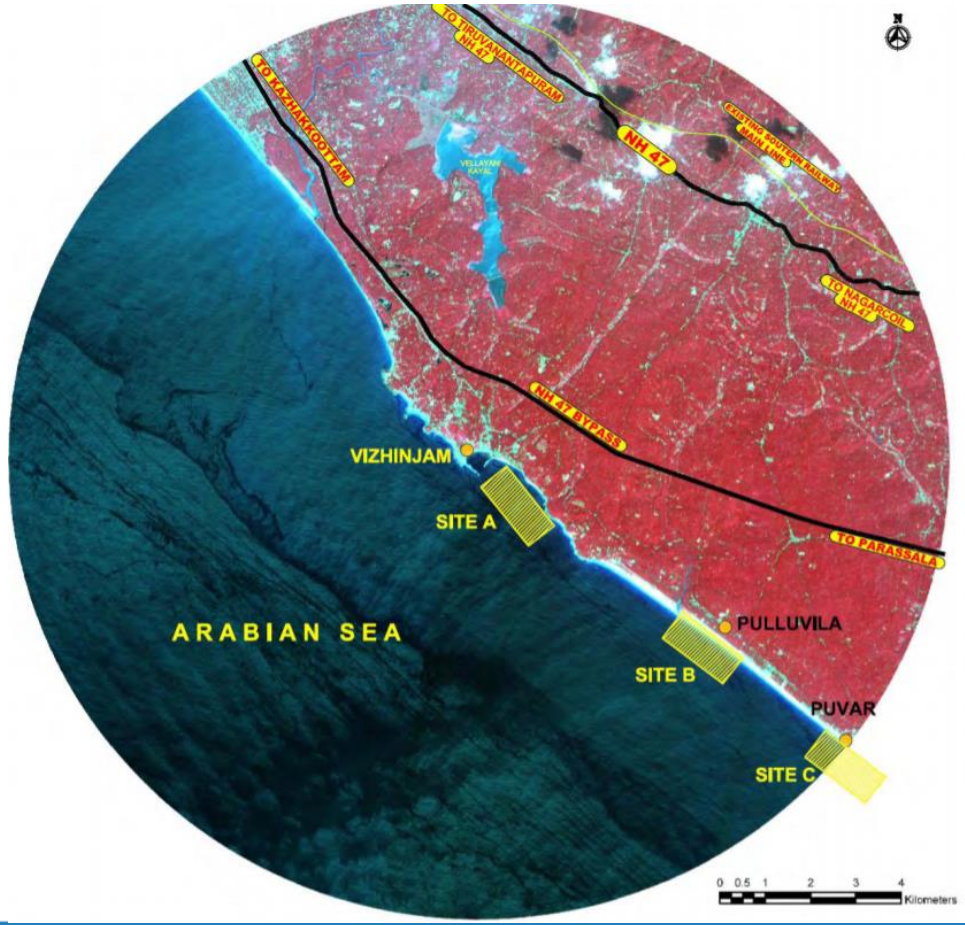
Comparison of alternative sites for port development

Site A	Site B	Site C
• About 3.5 km is a rocky coast with practically no continuous sandy beaches • Behind the coast, the steep rocky slopes rise towards a hillock area of 60-70 m height • Natural depth of 20 - 23 m is available at approximately 2.0 km from the coastline • Protection is available naturally by the presence of fishing harbor north of the site	• Approximately 2.0 km length with 400 to 500 m wide flat foreland in front of a steep laterite face up to a 50-60 m high hillock area • Area is frequently flooded during rains and is not inhabited, but filled with coconut trees • Natural depth of 16 - 20 m is available at approximately 2.0 km from the coastline	• Large low-lying area with marshy parts, prone to flooding from the river, backwaters and small hillocks • Natural depth of 20 m is available only at 2 - 3 km from the coastline. • Sandy foreshore is only approximately 50 m, and a rubble mound seawall is protecting Pozhiyoor village

Comparison of alternative sites for connectivity

Site A	Site B	Site C
• Presence of cliff behind the coastline poses difficulty for surface linkage access to the site • Alternative could be to make an access from around the base of rocky promontory to Adimulathurai village and from there to District road	• Village road connecting the District Road at Adimulathurai can access the proposed site • Widening of this road may be needed to cater to road traffic	• Access to the project is from the Poovar village through a single lane road • This site is connected to NH 47 through a two-lane road till Neyyatinkara village

Comparative access to highway for site alternatives



Satellite Imagery showing location of alternative sites with nearby national highway (NH-47)

Comparison of alternative sites for operation & navigation

Site A	Site B	Site C
• 200 Ha will be established by reclamation • Approximately 2.0 km from the coastline having a natural depth of 20-23 m. • Outer breakwater will be located at water depths of around 18 to 23 m • Breakwater is planned parallel to the coast based on the studies and it was found to be efficient.	• 150 Ha in the full development by reclamation. • Approximately 2.0 km from the coastline a natural depth of 16-20 m • Outer breakwater will be located at around 20 m water depth. • Breakwater layout is considered to be perpendicular to the coast to reduce siltation along the approach channel.	• 150 Ha in the full development by reclamation. • Approximately 2 - 3 km from the coastline is having a natural depth of around 20 m. • Outer breakwater will be located at 20 m water depth. • Breakwater layout is considered to be perpendicular to the coast

Comparison of alternative sites for environment

Site A	Site B	Site C
• Shoreline in this stretch is rocky characterized with thick laterite capping with exposures of gneissic rocks at the foothills • The beachfront in this stretch is nonexistent or very narrow at few locations • Environmentally sensitive coastal vegetation like mangroves is not observed along this stretch	• Coastal vegetation like mangroves is not observed in this stretch	• Backup area (swampy area) earmarked for the port is devoid of mangroves • The vegetation observed in the site is mostly represented by local floral varieties like coconut trees

Comparison of alternative sites for social aspect (Livelihood & tourism)

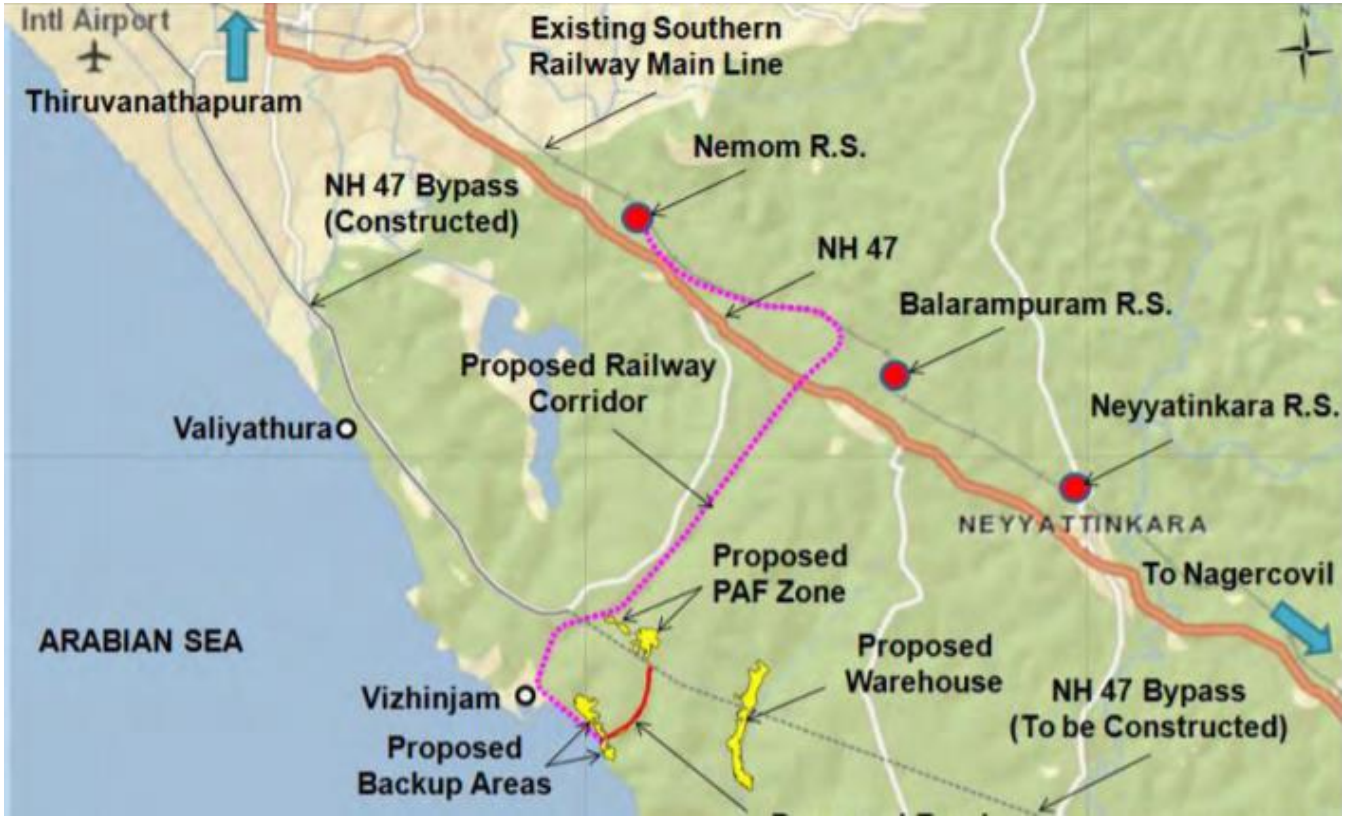
Site A	Site B	Site C
• Coastal stretch at this site shadow zone does not come under active fishing. Fishing activity in this area is confined to deep seas. • On the hill slopes overlooking the sea, several tourist resorts are located • Most likely attract a considerable number of cruise ships per year bringing thousands of tourists to the area	• Fishing activity is highly intensive in this stretch.	• There is no fishing activity in this beachfront. • Impact on the tourism front due to the development and operation of the port at this site (disturbing natural beauty of back waters)

Comparison of alternative sites for cost

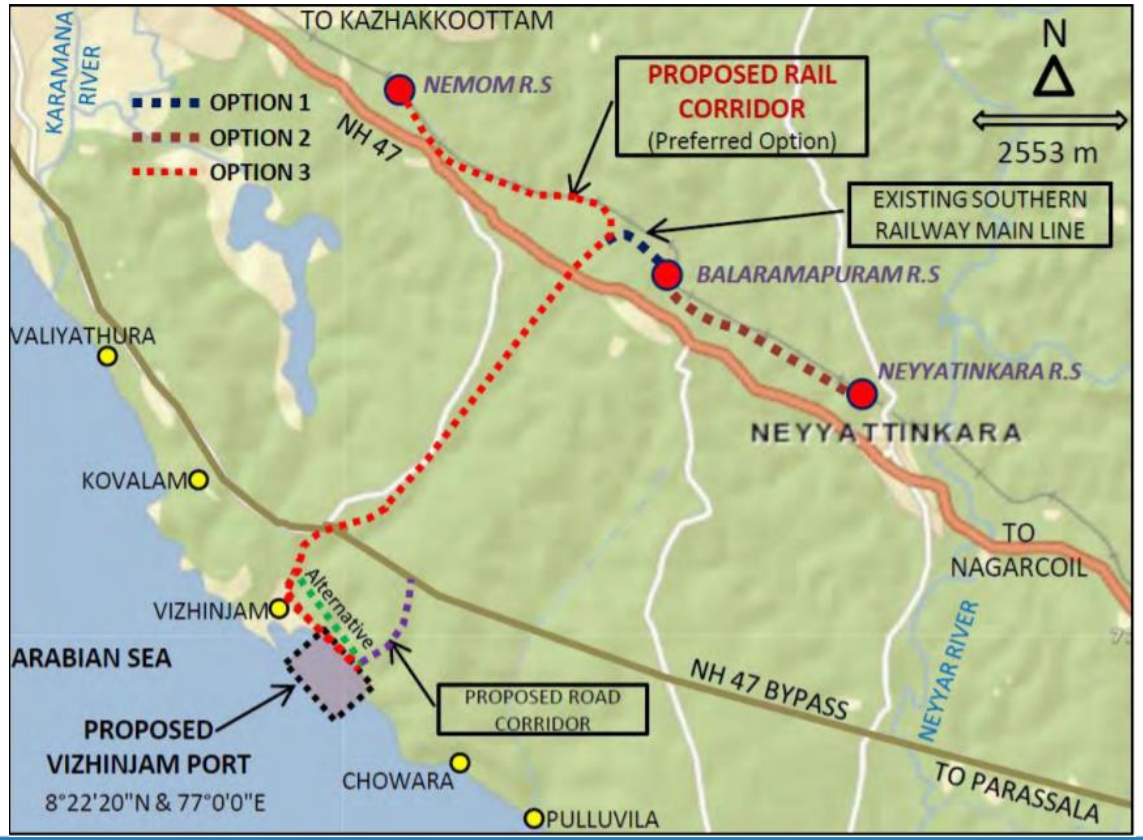
- Site A has the advantage of using cutting material along the land side for filling. Also, the breakwaters planned at Site A is flexible for phase wise construction
- Diversion of river may be taking place at Site B and Site C which will increase cost
- Maintenance dredging for Site A very minimal compared to Site B & C.
- All other facilities and infrastructure elements are assumed to be equal for the three sites

Based on comparative analysis of sites, site A was finalized for the project.

Road & rail connectivity with port

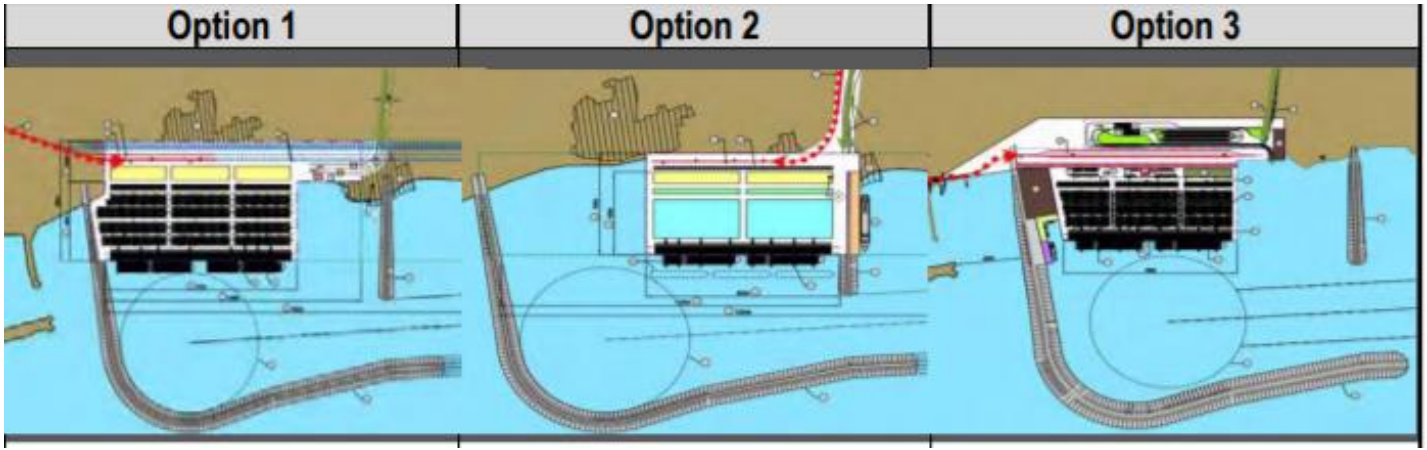


Rail line alternatives for project site



Alternatives for the proposed railway alignment

Alternatives for end-point terminal at Vizhinjam



Due to the limited options available for the rail and road access to the proposed port and significant cost involved in land acquisition, R&R, and technical issues for Option 1 and Option 2, Option 3 was considered and taken forward for the final master plan

Alternatives for end-point terminal at Vizhinjam

Planning Flexibility	**Moderate:** Flexibility in rail yard planning	**High:** highest level of flexibility in planning as the proposed road & rail connectivity would be expandable with port	**Low:** layout offers the lowest flexibility in planning due to the rail yard entry constraints because of "S" curve of the rail line
Social Impact	**High:** Social cost would be high due to the displacement, R&R of local community	**Medium:** Social cost would be overcoming the resistance of the local community due to the rail corridor physically dividing the land adjacent to the rail.	**Low:** Social cost would be minimal as the proposed rail corridor will be through the existing drainage so displacement of local community would be minimal
Technical Feasibility	**Difficult:** due to the unfavorable topography	**Very Difficult:** rail connectivity will need tunneling & land cut to overcome steep grading and highly unfavorable topography	**Moderate:** Technical difficulty to be overcome for the last mile connectivity (sharp "S" curve) which will be an elevated structure & yard entry has to be through the northern breakwater
Impact on Port Costing	**High** economic cost involved in Land Acquisition, R&R for the rail corridor	**High** economic cost involved due to the tunneling requirement for rail corridor	**Moderate** economic cost involved in providing an elevated structure for the last mile rail connectivity

Primary stakeholders for participation

- Project Displaced Persons (PDP)
- Project Affected Families (PAF)
- Project Affected Persons (PAP)
- Squatters Encroachers
- Project impacted communities
- PAFs in the road-rail connectivity alignment

Activities during public consultation
- Focus group discussion
- Informal discussion
- Interview
- Formal community discussions

Public participation: photo gallery



Exhibit 1: CEO of VISL discussing with District Collector



Exhibit 2: The Banner displayed on Public Hearing



Exhibit 3: A close-up of the crowd



Exhibit 4: Video and Still Camera

Media coverage of public participation

EIA hearing sees waves of support, criticism

Call for better compensation, housing projects for displaced people

Staff Reporter

THIRUVANANTHAPURAM: Support for the proposed Vizhinjam port, along with criticism that environmental concerns were not given a proper earmarked the public hearing on the crucial draft Environment Impact Assessment (EIA) for the project here on Saturday.

The event, held at the Aswathi auditorium at Vizhinjam, saw a large turnout of public representatives, members of non-governmental environmental organisations, representatives of the fishing community, and so on. Chaired by District Collector K.N. Satheesh, the public hearing organised by the State Pollution Control Board (PCB), witnessed public representatives cutting across political lines declaring their support for the project, but at the same time, calling for steps to ensure the rights of the local public.

Jameela Prakasham, Kovalam MLA, said it was a surprise that the local legislator was not included in the governing board of Vizhinjam International Seaport Limited (VISL) and demanded that amendments be made. Calling for better compensation packages for people whose land was acquired, housing projects for the displaced and a medical college in the region, she said the process had to be transparent and justified. Captain K. Gopakumar, Chief Staff Officer, Kochi Naval Base, said more than two lakh ships passed by the region every year and with threats like that of Somalian pirates existing in the area, Vizhinjam was crucial from the security point of view. The Indian Navy had for long wanted a port in the area and the proposed port was the right move in the direction, with the Navy being allowed a berth there. The multi-national presence in the Indian Ocean required such a facility for the Indian Navy for quicker response and reaction, he said.

The hearing was not devoid of commotions, with the first being during Ms. Prakasham's speech.

Her demand for more drinking water facilities in the region was met with slogans and then arguments, which snowballed into jostling between her supporters and the protestors, who alleged she was politicising the hearing. When T. Peter of the Kerala Swathantra Matsyathozhilali Federation came to present his views, he was met with another uproar, and though he presented his views, the hearing did not last long after that, being wound up by 3.30 p.m. C.V. Sundaraman, Head (Environment Section), L&T-Ramboll Consulting Engineers Limited, who made a presentation on the project, said suggestions and concerns submitted at the hearing would be incorporated in the final report.



FOR AND AGAINST: *District Collector K.N. Satheesh (right) looks on as an altercation broke out during a public hearing on the draft environment impact assessment for Vizhinjam port in the capital on Saturday.* – PHOTO: S. GOPAKUMAR

Tourism body backs project

THIRUVANANTHAPURAM: Even as hotels and resorts in the Kovalam region have aired their fears over the possible impact of the proposed Vizhinjam port on the tourism sector, the Confederation of Kerala Tourism Industry (CKTI) has thrown its weight behind the project, stating that the commissioning of the port is essential for the tourism industry in the State.

CKTI president E.M. Najeeb and secretary M.R. Sayanaban, in a statement submitted during the public hearing on the draft EIA report for the port at Vizhinjam here on Saturday, said the 'CKTI as a whole, supported the project and dreamt of its fulfilment.'

"The EIA report showed that between Nellikunnu and Admalathura up to the approach channel of the project and to the interiors, there were 31 resorts, out of which only 11 would be affected. Among these, there were only three big resorts having more than 10 rooms. Comparing the sacrifice of a few hotels at the location, with the dimension, size, and benefits of the entire project, we should, by all means, opt for the project only," they said.

Shoreline change fears misplaced, say experts

Staff Reporter

THIRUVANANTHAPURAM: Faced with mounting criticism that the Environment Impact Assessment (EIA) for the Vizhinjam port project, the draft report of which has been published, is a mere eyewash and that it does not address the concerns over possible shoreline changes, experts behind the report have said that the concerns are misplaced.

The concerns and the multiple aspects of the possible impact on the area were dealt with in detail during the assessment and made public in the EIA draft report, available on the Vizhinjam International Seaport Limited website: vizhinjamport.in. There were some quarters, who had read only the executive summary, and come out with allegations, they said.K.M. Sivakholundu, Scientist and Group Head, Coastal and Environmental Engineering, National Institute of Ocean Technology, and member of the VISL Technical Subcommittee, said the project site was not prone to severe erosion as was being made out. Different studies had proved this. At the same time, one suggestion, that of sand-bypassing, was just one among many possible techniques that could be employed to tackle whatever erosion that happened along the coast.



K.M. Sivakholundu and Rajesh P.R.

Rajesh P.R., head, numerical model studies, L&T-Ramboll Consulting Engineers Limited, said he was part of teams that had done similar EIAs for several ports, and the minimum impact so far seen was for Vizhinjam. The alternative site analysis, with Pulluvila and Poovar being the other sites considered, and the fact that Thiruvananthapuram had the most stable coast in the entire State, had supported the results of the shoreline studies.

"No significant change in the shoreline evolution pattern was observed towards the north-west side (along Kovalam) of the fishing harbour and at the Karamana river and further north of it. It may also be noted that the change in the shoreline evolution along the southeast side of the port will get stabilised over the years. However, it is recommended to monitor the shoreline during the development of each phase of the port," Mr. Rajesh quoted the Mathematical Study Report submitted as part of the EIA. "The entire report is available on the website for everyone to go through the details. That should address most of the concerns raised," Dr. Sivakholundu said.

Suggestions galore at public hearing for port project

Fishermen's Livelihood, Impact On Tourism Sector, Usage Of Water For Port Activities Among Concerns Raised

TIMES NEWS NETWORK

Thiruvanathapuram: The state pollution control board received around 200 suggestions and comments, a majority of them supporting the proposed Vizhinjam international seaport, during a public hearing conducted at Vizhinjam on Saturday.

Various citizen groups, members from fishing community, political representatives, activists, naval officials and project-affected stakeholders presented their views and concerns about the project. The mandatory public hearing was held as part of procuring the environmental clearance from the union ministry of environment and forest for the project.

A section of the fishing community expressed their concerns on the loss of livelihood and blockage of access to enter the seaway where the proposed project is coming up.

"We feel there are some issues that need to be addressed. We are not against the project, but we want to be assured that our livelihood will not be affected," said Jesurajan, president, Vizhinjam Fishermen Welfare Association.

Activists pointed out that massive usage of water for the port-related activities needs to be look at properly, especially when the area is facing huge drinking water shortage.

"The port will draw lakhs of litres of water from Vellayani Lake and studies need to be conducted looking at the cumulative holding capacity of the lake," Suresh, an advocate, said.

Earlier during the day, Vizhinjam port authorities made a detailed presentation on the proposed port based on the EIA report that has been submitted to the pollution control board.

"We had done a four-season study, which is now part of the EIA report. The port can attract 29 vessels passing in the Middle Eastern route and 64 vessels coming from the Suez Canal route every day," Sunder Rajan, head of environment division from L and T Ramboll said.

He said that around 40% of the vessels with Indian cargo land in Colombo, Singapore and Malaysia ports and once Vizhinjam becomes operational this huge cargo can be offloaded here.

"There will be negligible impact on tourism and the site selection was done after looking at 21 parameters," he said.

The Indian Navy said that the port is of strategic importance to the country given that there is a huge international cargo traffic presence. "We need berth to attack rafters and interceptors especially after concerns have been raised following the Mumbai terror attack and killing of innocent fishermen by Italian marines," Captain Gopakumar said.

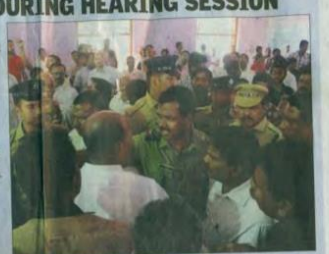
TURBULENT SCENES DURING HEARING SESSION

The public hearing for the proposed Vizhinjam international seaport presided by the district collector was disrupted twice. The first disruption occurred when local representatives of political parties and a section of project-affected stakeholders objected to certain remarks made by MLA Jameela Prakasham on drinking water supply and the second one occurred when Cyriac Kodath, who represented tourism-affected stakeholders, began his presentation.

There were also voices of protest when T Peter of Kerala Swathanthra Matsya Thozhilali Federation tried to raise some concerns about problems faced by the fishing community. Senior port officials said that it was unfortunate that some people tried to disrupt the proceedings as this would only hamper the mandatory process of public hearing.

"The public hearing is meant to provide an opportunity to individual stakeholders which is an integral part of democratic process. Disrupting the proceedings will not in anyway help in getting an eco-clearance instead, there should have been proactive interaction amongst various stakeholders," the official said.

Though some of the affected stakeholders were not allowed to speak and priority was given to politicians, they gave written representations expressing their grievances and comments. TNN



HIGH TIDE OF PROTEST: Police pacifying the stakeholders, who protested against a comment made by MLA Jameela Prakasham during the public hearing for the proposed Vizhinjam seaport project on Saturday

The tourism industry expressed concerns on the adverse impact on tourism, especially in the Kovalam stretch, which they said, would impact Rs 7,000-crore tourism revenue for the state.

"The port authorities should constitute an expert committee to monitor marine pollution and fish habitat in this stretch," Soman of Kerala tourism council said.

Former engineer with Vizhinjam port Appukuttan Pillai said that the final EIA report should also factor in concerns like cyclone probability as studies have reported such a tendency.

Locals' thumbs up for Vizhinjam

DC CORRESPONDENT
THIRUVANANTHAPURAM,
JUNE 29

According to the EIA report, eight hotels in the area would be affected by the project

Hundreds of locals favoured Vizhinjam International Transhipment Terminal at the public hearing on the Environmental Impact Assessment (EIA) draft based on the Pollution Control Board report. Local MLA Jameela Prakasham's comments on water shortage at Kovalam and also a local hotelier's call against the project disrupted the hour-long public hearing briefly.

On May 19, the PCB had published the draft EIA. The public hearing was conversed by district collector K.N. Satheesh which was followed by a presentation by L&T Ramboll environmental head C.G. Sunder Rajan. He explained the merits of the three-phased project with the first phase expected to cost ₹5,187 crore which will be completed by 2017.

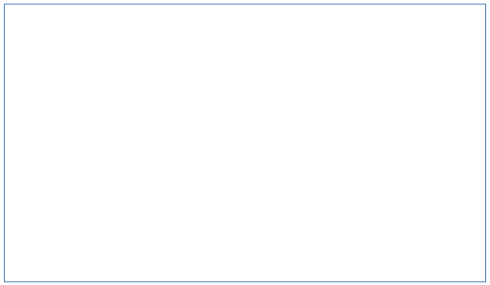
The proceedings were audio/video recorded and the PCB will submit a report to the ministry of environment forests (MoEF) with a copy to the project proponent. Vizhinjam International Seaport Limited (VISL). MoEF will then review the report and take a decision in 105 days.

With no land acquisition taking place on the second and third phase of the project, the landward area requirement for the first phase is 147 acres. Kovalam MLA Mr. Prakasham demanded that instead of the current five km for the area development study, 15 km should be included. He also demanded that the local MLA should be included on the board of VISL. Local journalist Elias John, who spoke in favour of the project, said that compared to Singapore port the tourism jobs offered are less.

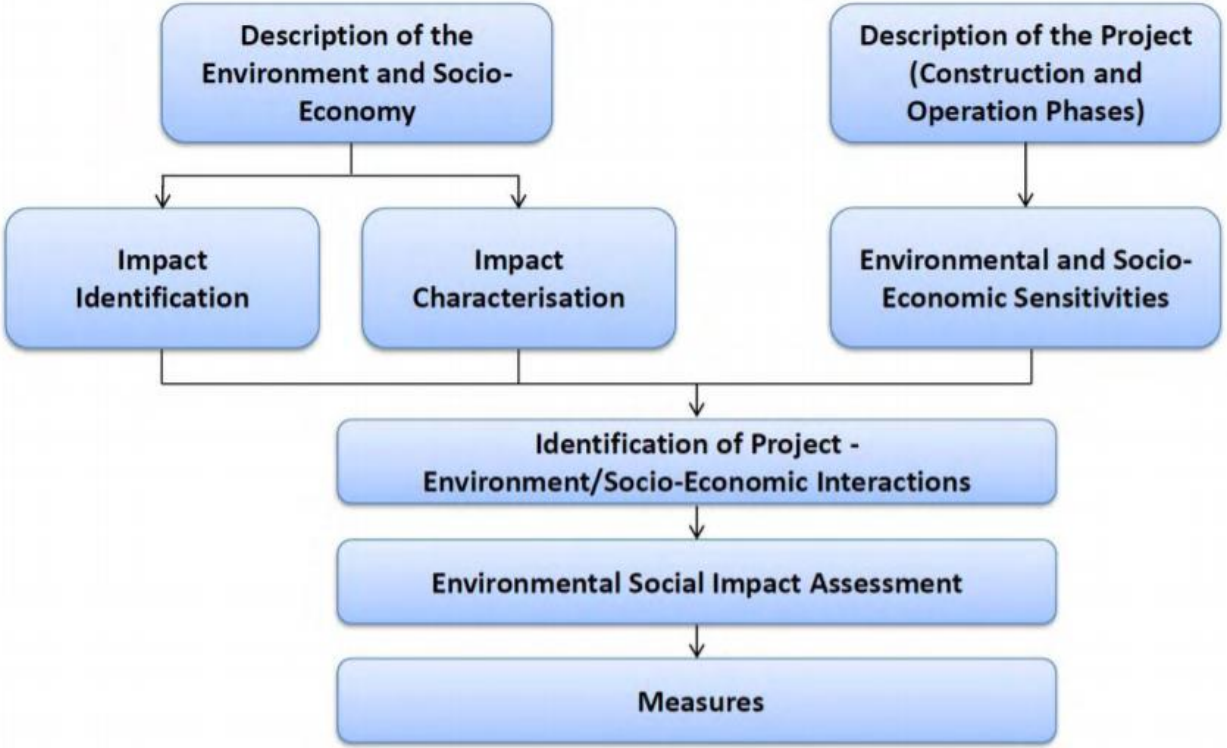
Coconut Bay resort owner Cyriac Kodoth, whose complaint had delayed the EIA by over 10 months, demanded that site selection should be shifted to Poovar, which witnessed disruption from the locals who had gathered at the venue.

Aspects of impact considered for project

- Land environment
- Water environment
- Marine environment
- Waste management
- Air environment
- Noise pollution
- Biological environment



Overview of impact assessment & mitigation process



Mitigation measures for change in land use

- The planning will be in accordance with landscape planning concepts to minimize major landscape changes.
- Land reclamation and change in land use pattern will be limited to the proposed port limits
- Project will be carried out in such a way that to ensure the proper drainage by providing surface drainage systems including storm water network, etc.

Mitigation measures for land reclamation impact

- Reclamation bunds and setting ponds shall be constructed
- Dredged material will be pumped into the reclamation area enclosed by reclamation bunds wherein the solids will be allowed to settle and the return water will be directed into sea through appropriate return channel/pipelines.
- The dredge fill will be covered by gravel before hard standing in order to prevent the seepage of return sea water into the groundwater,
- Suitable impervious liners such as LDPE will be provided all along the return water channel if required

Mitigation measures for impacts on shoreline

- Continuous monitoring of shoreline with the help of high-resolution satellite imageries during construction and operation phase.

- Fixing of shoreline just prior to construction phase by latest satellite imagery and then ground truthing on a regular basis with close monitoring and ground markings.

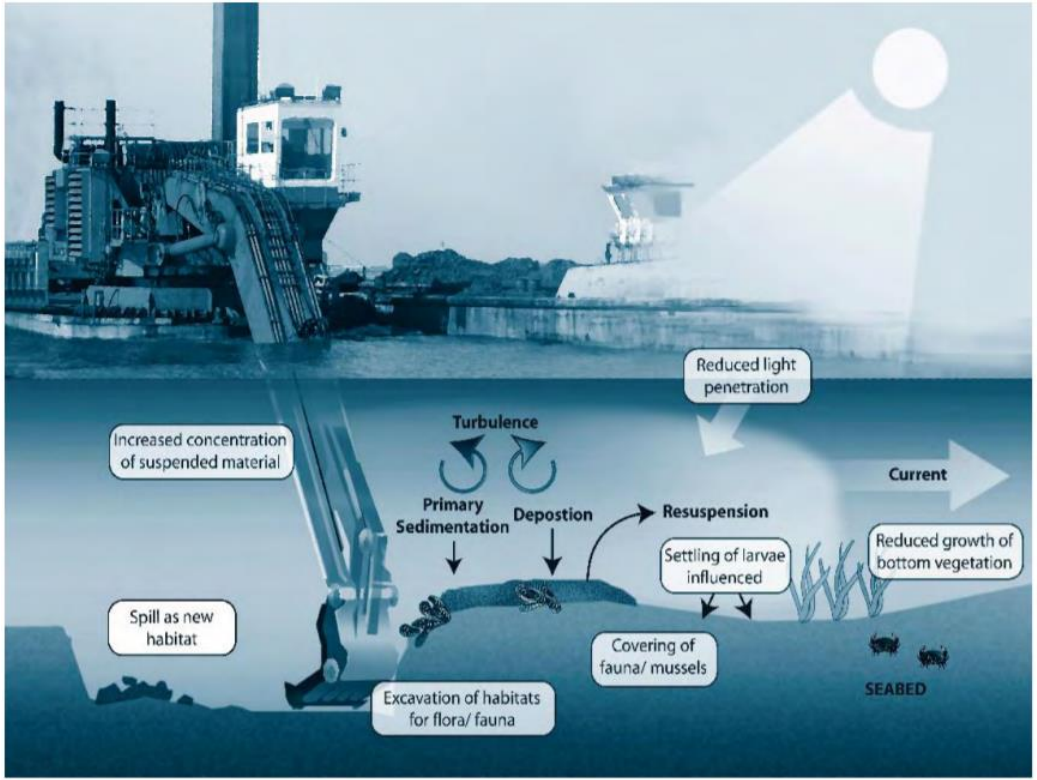
Mitigation measures for impacts on surface water

- Two polluting streams joining sea at new proposed fishing harbor site will be treated.
- Effective water conservation measure should be followed.
- Rain water harvesting/ storm water management in the projection region.
- Providing water & wastewater treatment facilities at construction camp.
- Regular monitoring of surface water for quality and quantity

Mitigation measures for impacts on water sources

- Baseline review of the existing groundwater resources and current consumption in the project region.
- A groundwater monitoring program should be implemented during the construction phase.
- When (deeper) aquifers are exploited the phreatic aquifer should not be affected.

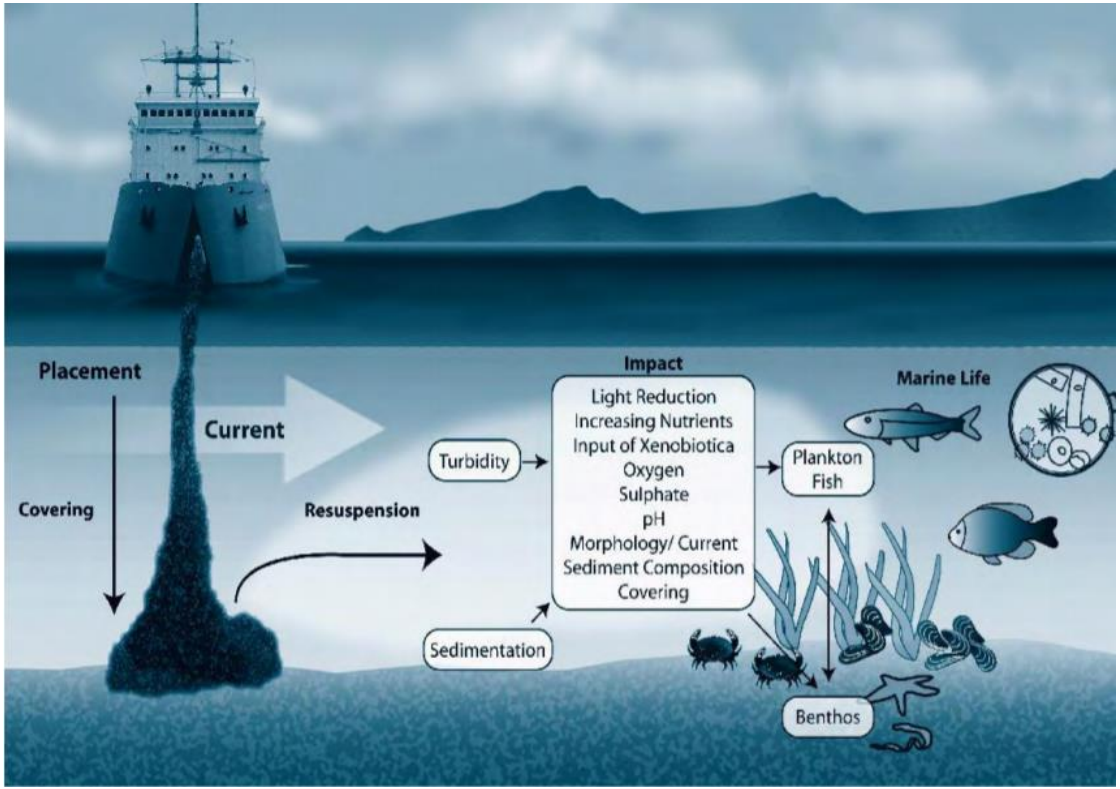
Impact due to capital dredging



Source: Axel and Christine, Terra et Aqua, No. 114, March 2009

- Increased concentration of suspended material
- Excavation of habitats for flora/fauna
- Reduced growth of bottom vegetation

Impacts due to dredged material disposal



- Increased turbidity in water
- Adverse impact on plankton fish

Source: Axel and Christine, Terra et Aqua, No. 114, March 2009

Mitigation measures for impacts of dredging

- Appropriate selection of equipment for pile driving and dredging
- Providing silt screen to avoid movement suspended solid for a longer distance
- Uniform disposal of dredged material at identified disposal location

Impact on green cover & mitigation measures

S. No.	Road Alignment			Rail Alignment			Growth of the Trees along Proposed Road/Rail Corridors
	Impacted Trees	Total Nos.	Area in Hectares	Impacted Trees	Total Nos.	Area in Hectares	
1	Coconut trees	1062	3.44	Coconut trees	6898	27.59	Fully grown and yielding
2	Plantain (Indigenous banana)	7140	2.55	Plantain (Indigenous banana)	13356	4.77	Seasonal Two times a year
3	Tapioca crop	1140	0.57	Tapioca crop	4600	2.30	Seasonal two times a year
4	Paddy	-	1.05	Paddy	-	9.63	Two times a year
5	Other trees Mango, Jack fruit, Tamarind,	100	0.50	Other trees Mango, Jack fruit, Tamarind,	1000	3.00	Perennial crop
		Total	**8.11**		**Total**	**47.29**	

As a part of total loss of green cover VISL is committed to plan about 25-hectare vegetation cover along the acquired land

Sources of air pollution during construction

- Material sourcing at Quarry Material transport
- Equipment transport
- Stone crushing operations in the crushers
- Handling and storage of construction material
- Construction and allied activities.

No major air pollution source during operation phase of project.

Mitigation measures for air pollution

- Material transportation through sea will reduce dust & emissions
- DG sets should be used only during power cuts
- Low Sulphur content diesel shall be used as fuel for generator sets
- Vehicles older than 15 years will not be allowed to operate during construction phase
- Storage facilities shall be equipped with leak detection systems.
- Vegetation screens alongside the port access roads

Mitigation measures for noise pollution

- Acoustic fencing might be installed along the edge of the access road
- Vehicles and generator sets to be serviced regularly and maintained properly to avoid any unwanted generation of noise or vibration
- Employees working in noisy environment should be made to wear earmuffs/ear plugs to avoid any adverse impact of noise

Mitigation measures for solid waste

- Solid waste shall be adequately collected and managed
- Appropriate facilities shall be provided for the reception of all wastes arising from ships.
- These should include facilities for the following basic categories of ship-generated wastes:
  - Oily waste (usually oil mixed with larger quantities of seawater, also fuel residues and sludge)
  - Garbage (originating from crew and passengers, maintenance of the ship, cargo and fishing activities)

Mitigation measure for impacts on livelihood

- Additional fish landing space to ease the congestion at the existing fishing harbor
- Improve the living conditions of economically displaced
- Culturally sensitive and economically sustainable income restoration measures
- Identify and provide special assistance to people who are especially vulnerable to economic displacement impacts

Monitoring programme for project

Attribute	No. of location
Air Quality	5
Noise level	5
Ground water	3
Surface water	3
Soil	4
Marine water	5
Plankton & Benthic	5
Sediments	5
Shoreline	15 km north & south (satellite)

Environment management plan (EMP): main attributes

- Water conservation measures
- Storm Water Management
- Effluent treatment & Disposal
- Spill Containment and Management
- Greenbelt/green areas Development
- Environmental monitoring program
- Institutional mechanism

Environment management plan (EMP)

- Rs. 400 Million (Rs. 40 Crores) has been earmarked for EMP measures
- "ZERO TOLERANCE FOR POLLUTION" will be an institutional slogan during construction and operational phase of the project
- An elaborate decentralized solid waste management facility
- Rainwater harvesting programme and other water resources development initiative

Conclusion

- Proposed port will attract commercial cargo traffic from international sea routes passing from near by as well as support coast guard and navy.
- Detailed study has been done to assess the various possible impacts of project on local environment including sea and marine life in nearby area.
- Mitigation measures are adopted and environment management plan and monitoring plan are in place to implement and evaluate these measures .

References:

- Vizhinjam International Seaport Limited, Government of Kerala; 2013 “Comprehensive EIA for Vizhinjam international deep-water multipurpose seaport”

Thank You